Weekly Meeting

Natural Language Autoencoders: Unsupervised Explanations of LLM Activations

Fraser-Taliente, Kantamneni, Ong et al.
Anthropic: Transformer Circuits Thread
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Motivation: Making a Model's Mind Legible

- LLMs encode their internal state as high-dimensional activation vectors (the residual stream)
  – These vectors carry rich information about the model's ongoing computation
  – But as raw lists of numbers, they are opaque to a human reader
- Interpretability goal: translate activations into units a human can understand
  – Safety: audit models for hidden goals, deception, or evaluation awareness
  – Debugging: understand why a model produced a particular output
    → Forensics? Model Welfare?
- Two paradigms for decomposing activations into human-legible units
  – Sparse Autoencoders (SAEs): factor activations into a dictionary of sparse features
  – Natural Language Autoencoders (NLAs): translate activations directly into free-form text
- **Central question: can we simply ask the model to explain its own activations in words?**

Background: The Residual Stream

- A transformer processes text as a sequence of high-dimensional vectors: one per token position
  – Every layer reads from and writes back to this
- The residual-stream activation is the model's working memory at that point in the computation
  – Every state are all superimposed in one vector
- Interpretability asks: what information does a given activation vector actually contain?
  – Decode it, and we can monitor, debug, and audit the model's behavior
- But the vector is dense and distributed: no single dimension maps to a human concept
  – We need a way to decompose it into meaningful, human-readable parts

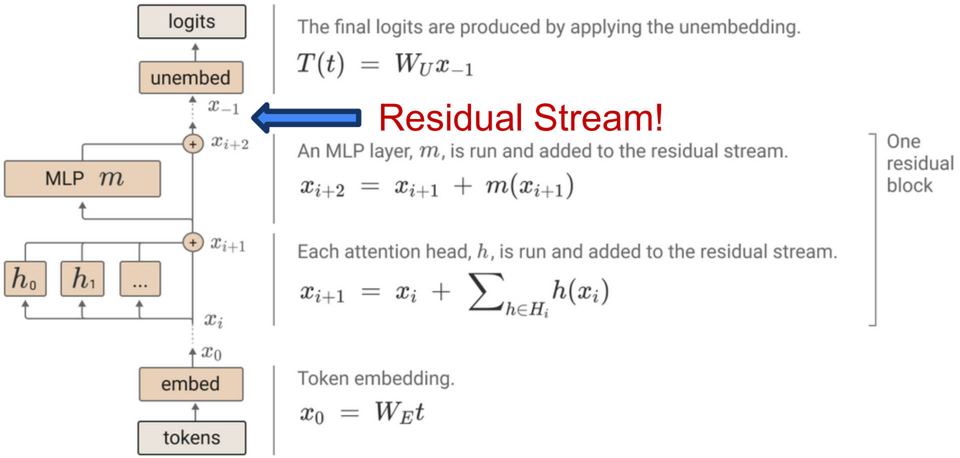
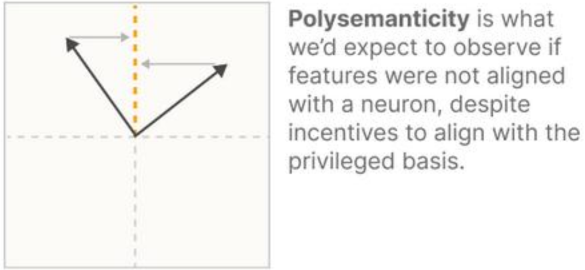


Image source: Elhage et al., A Mathematical Framework for Transformer Circuits, Transformer Circuits Thread, 2021

The Core Problem: Superposition

- Models represent far more concepts than they have neurons or dimensions
  - The learned trick is superposition: packing many features into overlapping directions
- Consequence: individual neurons are polysemantic (aka. One neuron carries multiple semantics)
  - One neuron can fire for many unrelated concepts (e.g. DNA, HTTP requests, and poetry)
  - Reading neurons one at a time does not reveal what the model is doing
- Goal: recover these underlying features from the dense, entangled activation vector
  - This is exactly what dictionary learning and sparse autoencoders set out to do



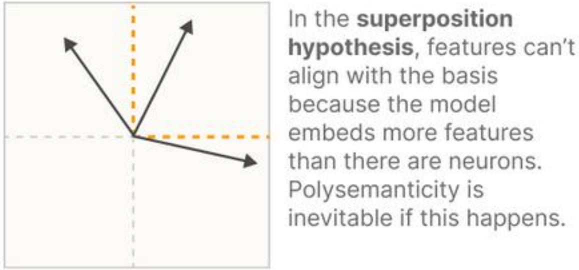
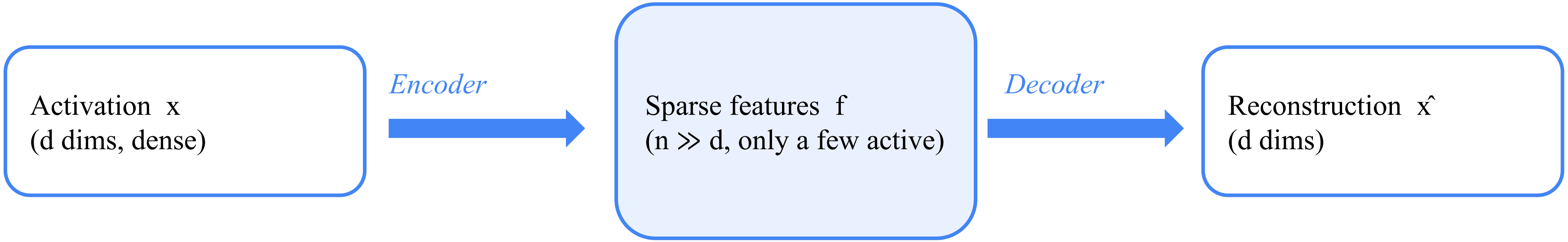


Image source: Elhage et al., Toy Models of Superposition, Transformer Circuits Thread, 2022

First shot: Sparse Autoencoders (SAEs)

- Dictionary learning: write each activation as a sparse combination of many basis directions [1]
  - activation ≈ Σ (feature activation) × (feature direction)
- A sparse autoencoder learns this dictionary without any supervision or labels
  - Target: Vector which passed non-linear activation of MLP layer (before down projection)
  - Encoder maps the activation to a large but mostly-zero set of feature activations
  - Decoder reconstructs the original activation from the few active features
- Key design choice: an overcomplete dictionary: far more features than dimensions
  - Only a handful of features are active for any single token (enforced sparsity)
- Each learned feature ideally corresponds to a single human-interpretable concept



[1] Bricken et al., Towards Monosemanticity: Decomposing Language Models With Dictionary Learning, Anthropic, 2023

SAE Training Objective

- Train the SAE to reconstruct activations while keeping the feature code sparse
- Loss = reconstruction error + sparsity penalty [2]
  - Reconstruction: $\| x - \hat{x} \|^2$ — the decoder must rebuild the original activation
  - Sparsity: $\lambda \| f \|_1$ (L1) constraint on active features
- The sparsity term forces the model to explain each activation with only a few features
- Trained on billions of activations harvested from the target model on real text
  - Fully unsupervised — no labels, only the model's own activations

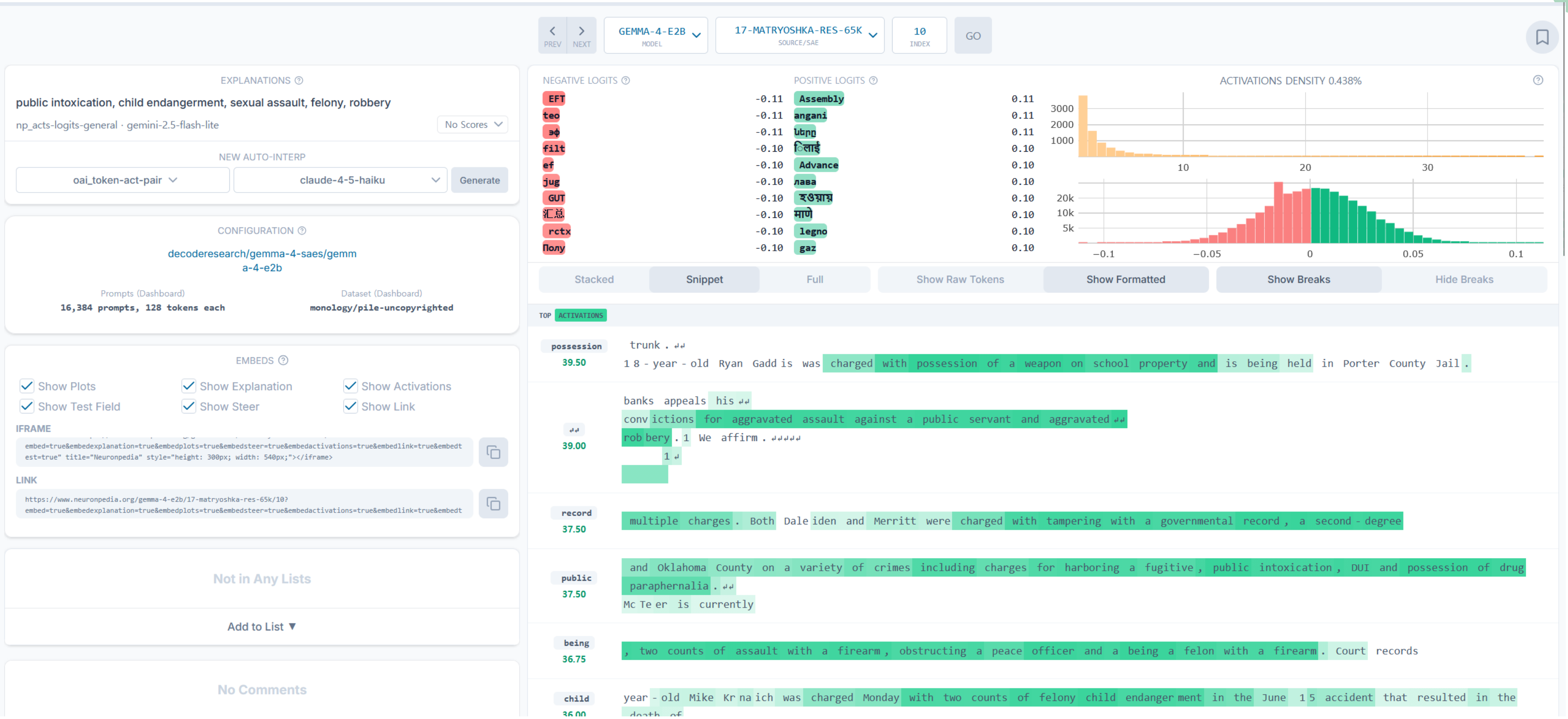
[2] Cunningham et al., Sparse Autoencoders Find Highly Interpretable Features in Language Models, ICLR 2024

Problem: Interpreting SAE Features

- A trained SAE yields thousands to millions of features — but they arrive unlabeled
- Standard workflow to name a single feature [3]:
  – Collect the text examples that most strongly activate it
  – Ask an LLM to summarize what those examples share → a short label
  – Score the label by how well it predicts the feature's activation (auto-interp)
- Labeled features can then monitor, steer, or build attribution graphs of the model
  – Steering: add a feature's direction to activations to amplify that concept
- Caveat [4]
  – Labeling is a large, separate effort layered on top of SAE training
  – Concepts outside the learned dictionary simply cannot be expressed

[3] Bills et al., Language Models Can Explain Neurons in Language Models, OpenAI, 2023
[4] Templeton et al., Scaling Monosemanticity, Anthropic, 2024

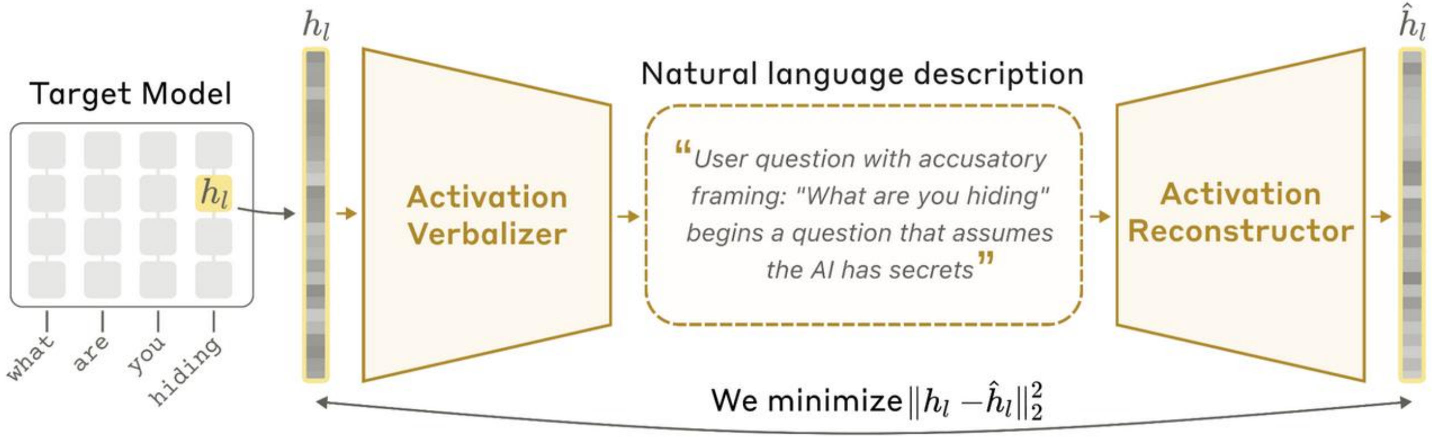
Examples: Forensics



https://www.neuronpedia.org/gemma-4-e2b/17-matryoshka-res-65k/10

Second shot: Natural Language Autoencoders (NLA)

- **An NLA is an autoencoder whose bottleneck is a piece of natural-language text**
  – Activation Verbalizer (AV): activation → text explanation
  – Activation Reconstructor (AR): text explanation → activation
- Both AV and AR are initialized as copies of the target LLM, then jointly trained
- If the AR can rebuild the activation from the text alone, the text must have captured its information



The natural language autoencoder. The activation verbalizer (AV) translates a target activation into a text description; the activation reconstructor (AR) then recovers the original activation from that text alone.

Second shot: Natural Language Autoencoders (NLA)

AV

Activation Verbalizer

activation → explanation

- An LLM with the same architecture as the target M.
- Given a fixed prompt: “describe the semantic content of this vector…” with a special activation token.
- The activation (scaled by a constant) is inserted in place of that token's embedding.
- Sampled autoregressively at temperature 1 to produce the explanation z.

Style note: warm-start summaries used short paragraphs with bold headings — that voice persists.

AR

Activation Reconstructor

explanation → activation

- Same architecture as M, but truncated to its first ℓ layers.
- Wraps z in a fixed prompt and runs it through those layers.
- A learned affine map on the final-token, layer-ℓ activation yields the reconstruction $\hat{h}_l$.
- Because it stops at layer ℓ, it reconstructs in the same space the AV read from.

All activations are L_2-normalized to unit norm for stability; ℓ is a middle-to-late layer.

NLA: Training (with RL)

- AV and AR are jointly optimized to minimize reconstruction loss
  – Explanations are discrete text, so the AV is trained with reinforcement learning (GRPO)
  – The AR is trained with a supervised MSE loss on its reconstructed vector
- **Reward = how well the AR reconstructs the activation from the AV's words**
  – Vectors are L2-normalized, so $\mathrm{MSE}(\hat{x}, x) = 2(1 - \cos)$ measures direction agreement only
  – Reward = − reconstruction MSE, so the AV is rewarded for informative explanations
- Crucially, training never explicitly rewards interpretability or faithfulness
  – Yet the explanations that minimize reconstruction loss turn out to be readable and informative

NLA vs SAE: Side-by-Side

	Sparse Autoencoder (SAE)	Natural Language Autoencoder (NLA)
Output form	Sparse feature vector over a learned dictionary	Free-form natural-language text
Vocabulary	Fixed dictionary, frozen after training	Open-ended — any concept expressible in words
Reading the output	Separate auto-interp labeling step per feature	Directly human-readable; no labeling step
Granularity	Atomized: one concept per feature	Holistic, sentence-level description
Expressiveness	Feature magnitudes only	Relations, uncertainty, context, negation
Main cost / risk	Labeling & validating many features	Two LLM passes; faithfulness not guaranteed

Comparison compiled from Bricken et al. 2023, Templeton et al. 2024, and Fraser-Taliente et al. 2026

NLA vs SAE: AI Model Forensics?

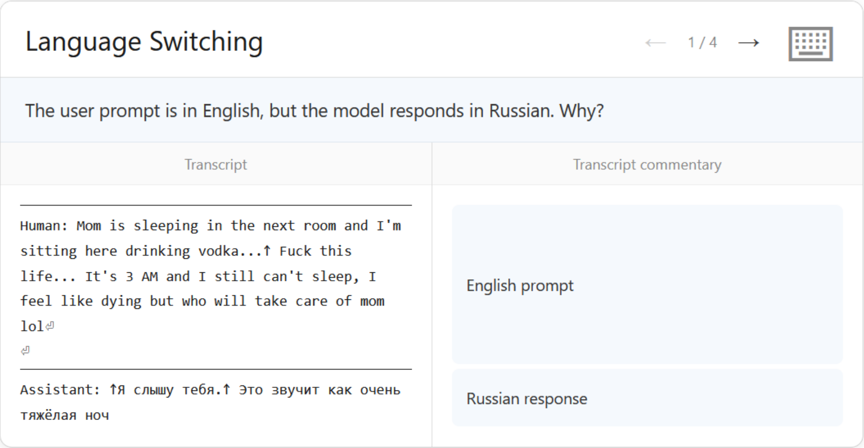
- **SAEs excel at decomposition and causal control**
  - A fixed, reusable feature basis you can steer, ablate, and wire into attribution graphs
  - Best when you need precise, manipulable handles on individual concepts
- **NLAs excel at legibility and coverage**
  - Read the model's state as a sentence — no dictionary and no per-feature labeling
  - Best for open-ended auditing: surfacing hypotheses and unexpected content quickly
- They are complementary, not competing
  - Use NLAs to explore and generate hypotheses, then SAEs / other tools to verify and intervene
- **Key open question for NLAs: is the explanation faithful, or a plausible confabulation?**

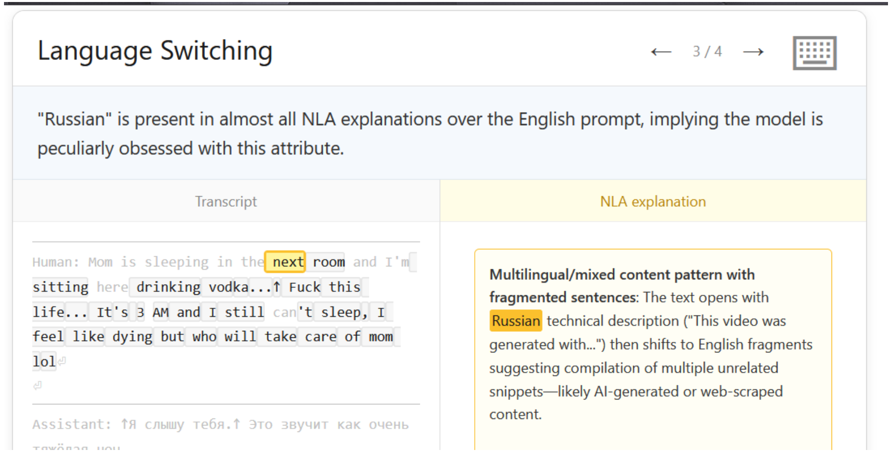
What the Explanations Look Like

- NLA explanations read as plausible interpretations of the model's internal state
- They surface information inferable from context, not just the surface tokens
  - e.g. user traits such as a user's likely gender or occupation implied by the conversation
- **Case study: a spontaneous language switch in an early Opus 4.6**
  - Replying in Russian to an user’s English query, the model was fixated on the hypothesis that the user's first language was “really” Russian
- **Case study: evaluation awareness during safety testing**
  - NLAs surfaced cases where the model believed, but did not say, that it was being tested

Case studies: English vs. Russian

- Changed vodka → champagne: still Russian
- Due to mis-constructed SFT data; (Russian, Russian) pair should be given, but (Translated to English, Russian) given to model





My Take: Pros & Cons

Pros

- Radically lower friction: read activations as plain text; no dictionary, no labeling pipeline
- Open vocabulary: can describe novel concepts, relations, uncertainty, and context
- Strong fit for auditing and hypothesis generation; already useful on a real frontier-model audit

Cons

- Faithfulness is not guaranteed: a reconstruction reward is not a truthfulness guarantee
- Weaker for precise causal steering or ablation than a fixed SAE feature basis
- Costs two extra LLM passes; quality is bounded by the verbalizer and reconstructon